AIAS+ 2026 · ORAL PRESENTATION

FlyGM

Whole-Brain Connectomic Graph Model for Whole-Body Locomotion Control

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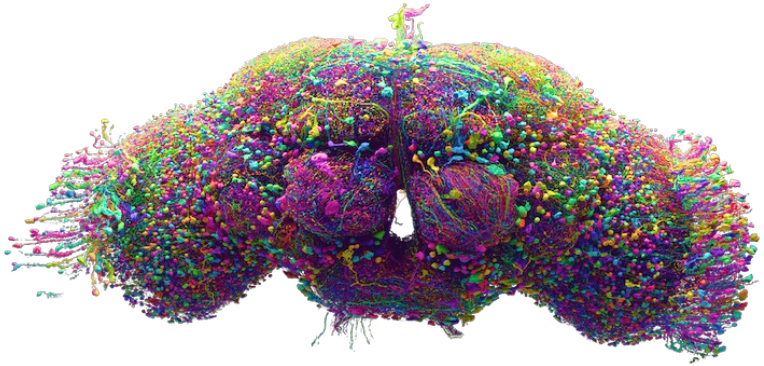
Demo page

WHY NOW

Fly brain and body are now both digital



139,255 neurons

54.5 M synapses

8,453 cell types

2024 — **FlyWire whole-brain map**
Complete adult brain wiring [2, 3]

2024 — **Connectome-based models**
Wiring predicts neural activity [4, 5]

2025 — **flybody simulator**
A whole fly body in physics [6, 7]

2026 — **Male CNS connectome**
Brain + nerve cord, 166,700 neurons [8]

Rendering: FlyWire [2]. Counts: [2, 3, 8].

2026: THE QUESTION GOES PUBLIC

Can a wiring diagram drive a body?

SEP 2025 · OURS

FlyGM paper drafted

Connectome drives a whole-body fly [1]

→

MAR 2026

“We’ve uploaded a fruit fly”

Eon Systems demo goes viral [9]

→

MAR 2026 →

“No, we haven’t uploaded a fly yet”

Neuroscientists push back [10, 11]

WHAT THE CRITICS POINT OUT

Synapse strength
Paths, not influence [12]

Neuron dynamics
Simplified neuron models [4]

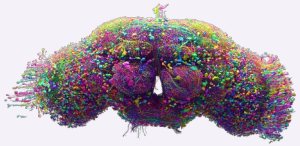
Body interface
Hand-picked control neurons [9, 11]

Open question: how should a connectome control a whole body?

FlyGM first draft completed 18 Sep 2025 [1]. Public discussion: [9–11].

OUR APPROACH

FlyGM: wiring as the controller's architecture



Emulate the wiring

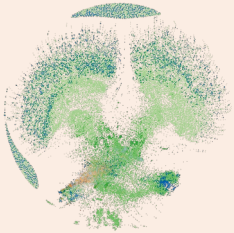
+ Real wiring, no training
− Body moved by built-in controllers

e.g. [4, 9]

Learn a generic net

+ Realistic walking and flight
− Network unrelated to the brain

e.g. flybody [7]



FlyGM (ours)

+ The real wiring is the network
+ Learns whole-body control

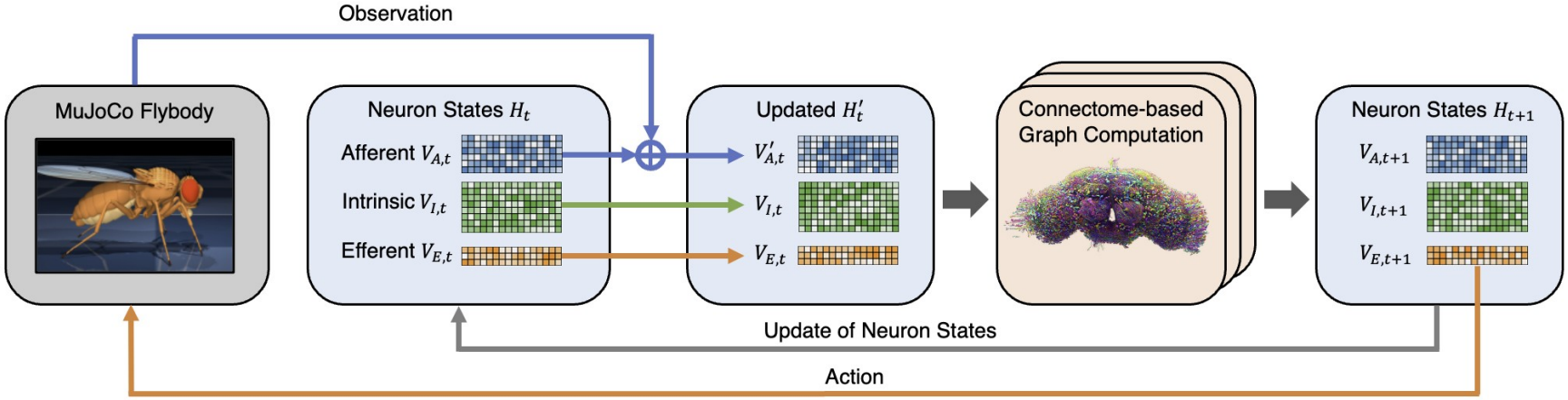
RESEARCH QUESTION

Can whole-brain wiring be a useful prior for whole-body control?

Images: FlyWire brain [2]; flybody fly [7]; FlyGM connectome layout [1].

METHOD

FlyGM closes the sensorimotor loop



1 Observe

Body senses → encoder → afferent neurons

2 Propagate

Signals flow along real synapses: M = W H

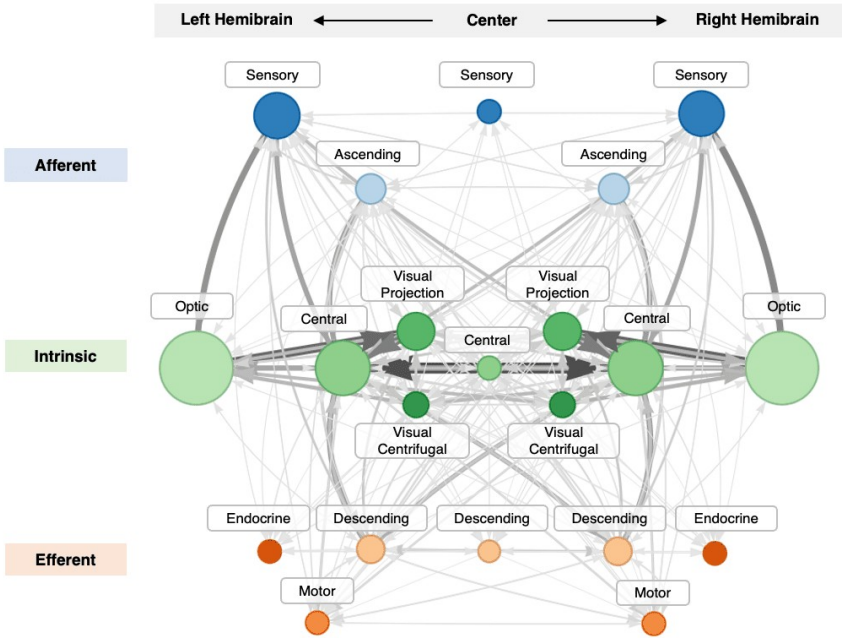
3 Act

Efferent neurons → decoder → 59 actuators

[1] Fig. 1, §2.1. Walking: 1,253-D input, 59 actions; flight: 104-D input, 12 actions.

METHOD

Wiring is fixed; computation is learned



FlyWire neurons grouped by class [1, 2]

FIXED from the connectome

- Who connects to whom
- Signed weights from synapses [13]
- Afferent → intrinsic → efferent

LEARNED by training

- Encoder and decoder
- A descriptor per neuron, η_v
- One shared update network, f_ψ

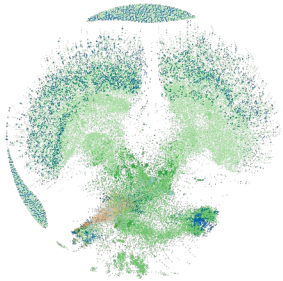
$$M_t = W H_t \qquad H_{t+1}[v] = f_\psi(M_t[v] \,\|\, \eta_v)$$

[1] §2.1. States are learned vectors, not firing rates; signs follow [12, 13].

TRAINING

Imitation gives a head start, PPO refines

1



2

Expert
flybody MLP [7]

Imitation
Match the expert's actions

FlyGM
Initialized policy

PPO
Maximize task reward [15]

Tasks
Walk, turn, fly

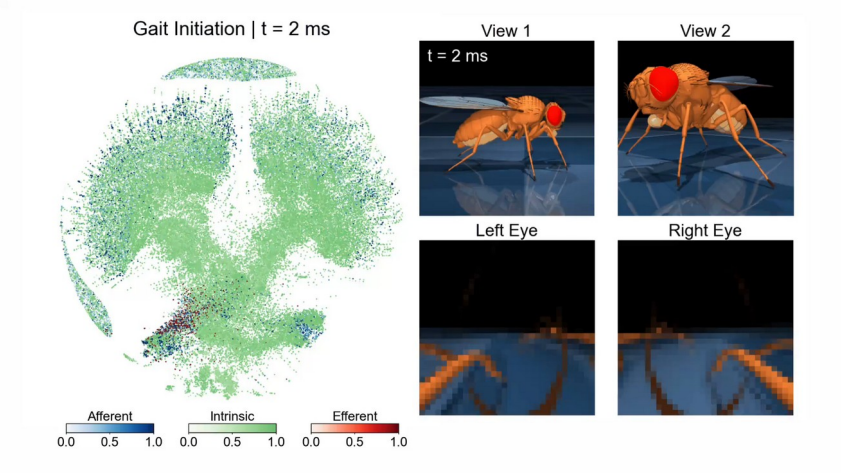
Same pipeline for every controller compared

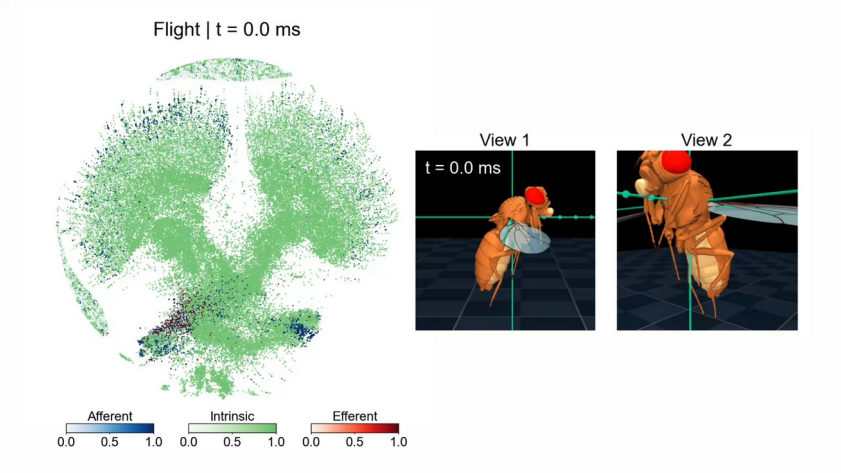
Walking and flight: separate policies

[1] §2.2, Appendix B. Expert policies and environments: flybody [7].

BEHAVIOR

Coordinated walking, turning and flight





Walking policy

Onset ≈80 ms · **Tripod gait** 3 cm/s · **Turns** 10 rad/s

Flight policy separately trained

Forward flight 20 cm/s · **Wing-beat control**

Click a video to play. Playback is slowed; read the simulation timestamps.

[1] §3.2, Figs. 3–4. Left panels: connectome layout with afferent, intrinsic and efferent activity.

Real wiring learns faster, steers better

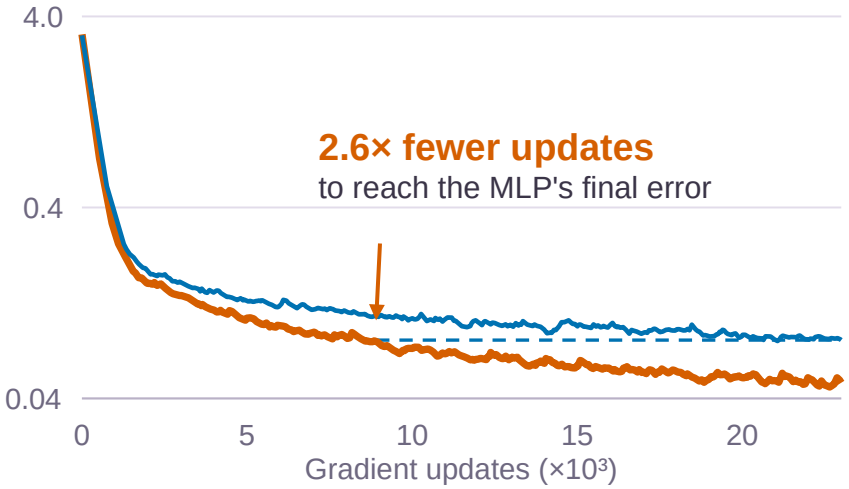
FlyGM
Real wiring

DP-Rewired
Same degrees, shuffled

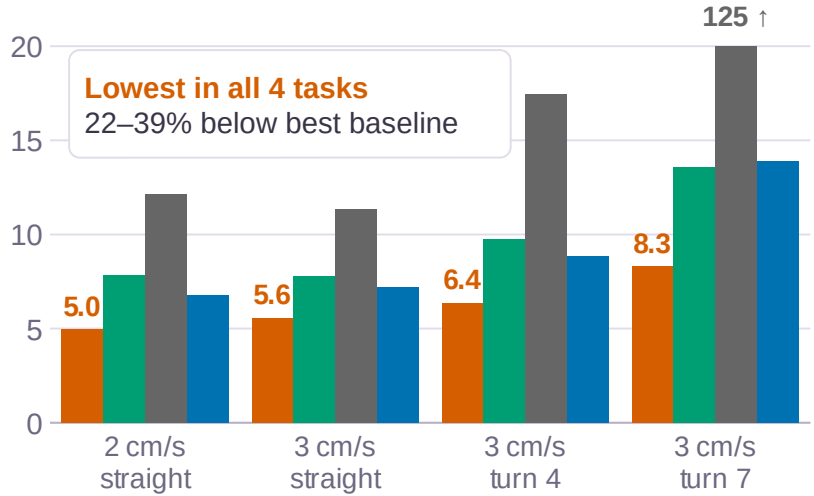
ER-Random
Same size, random

MLP
No graph, more params

Imitation: error to the expert (log)



After IL + PPO: angle error (°)

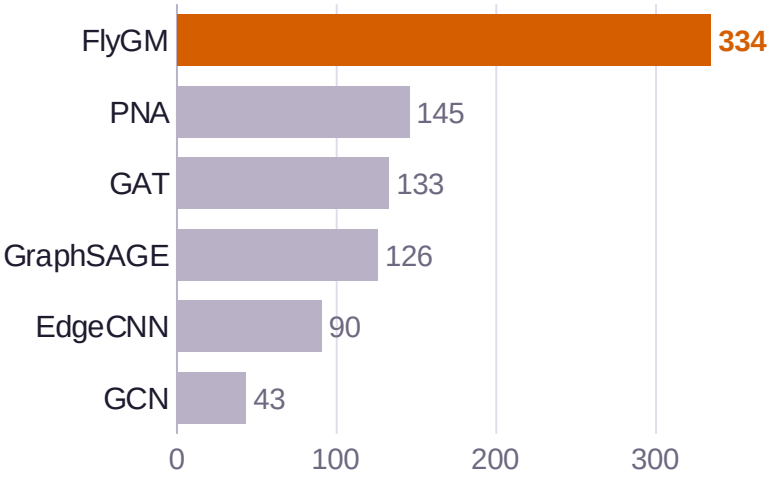


[1] Fig. 2a (one trace per model) and Table 1 (10 rollouts per condition; turn in rad/s).

The gain survives ablations and GNNs

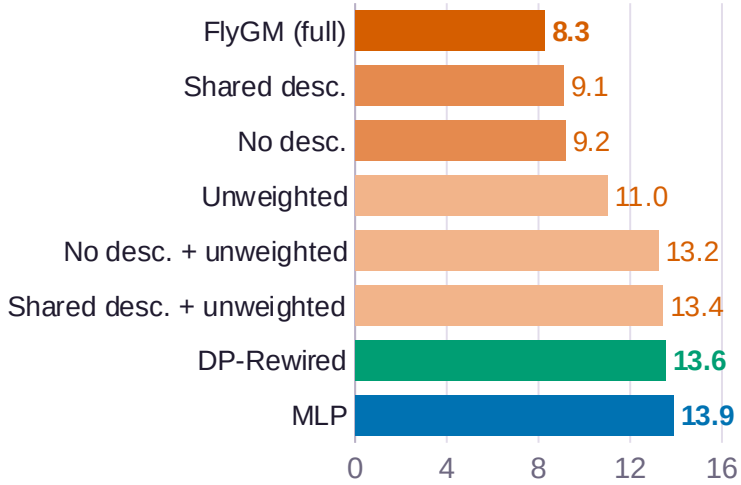
Generic GNN layers fall short

Same FlyWire neurons; reward after IL + PPO



Every connectome variant wins

Angle error (°) at 3 cm/s, turn 7 rad/s

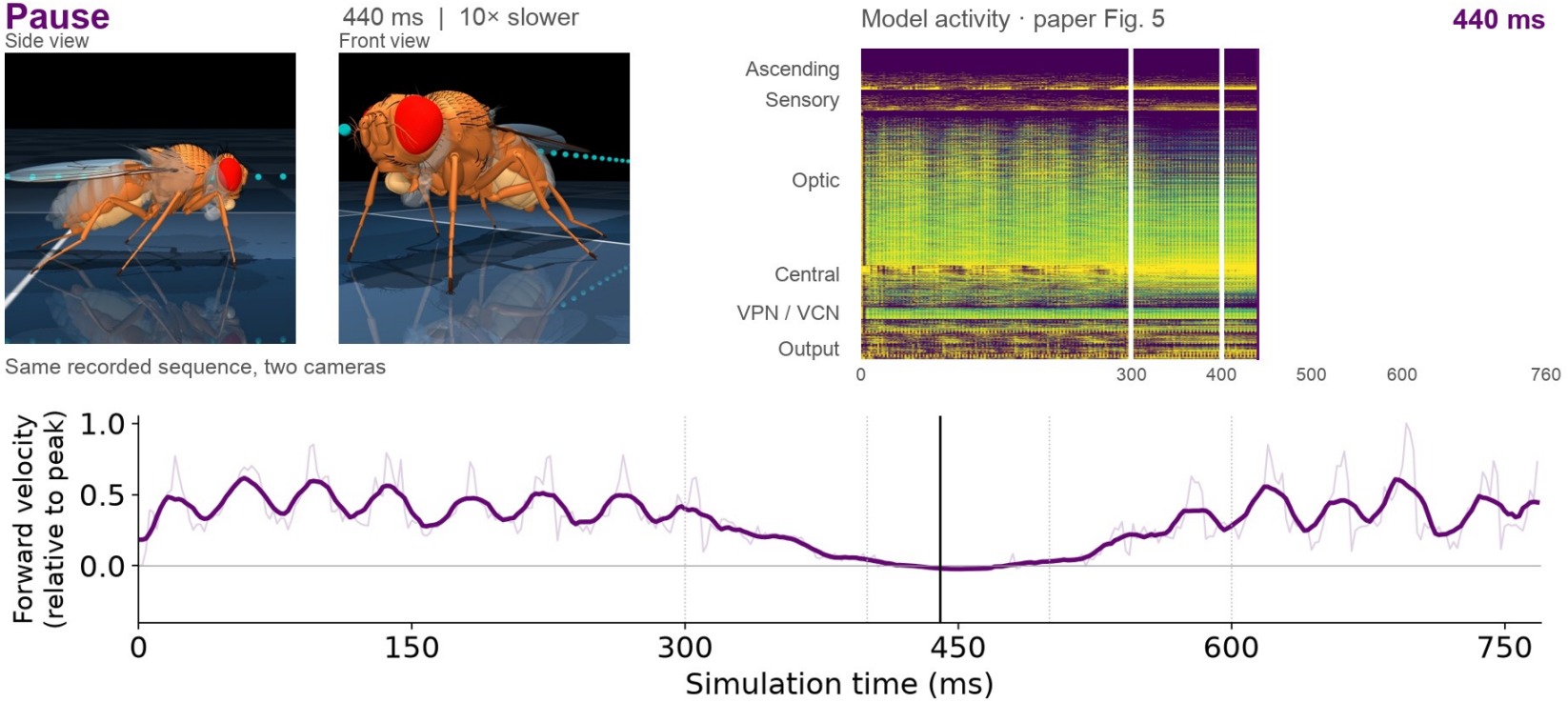


2.3× the reward of the best GNN (PNA)

Signed weights help most: 8.3° vs 11.0°

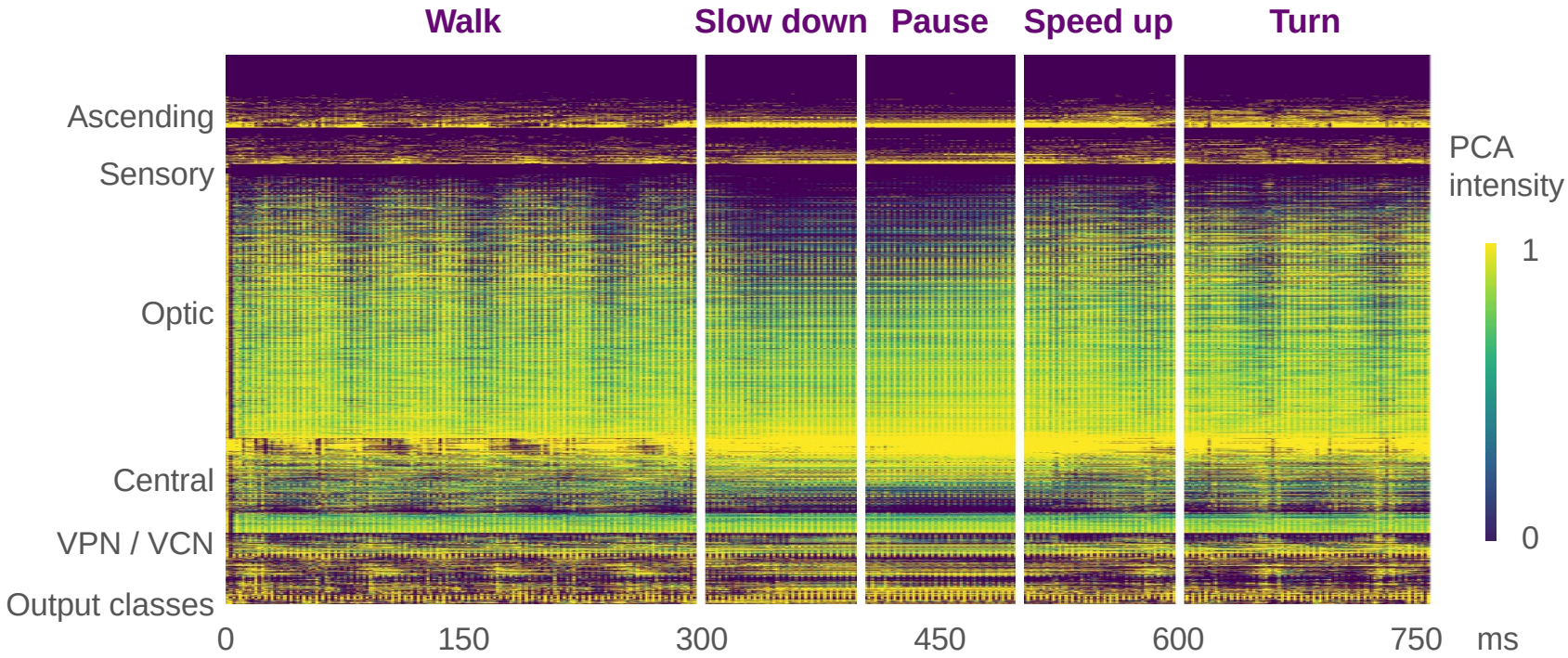
[1] Tables 2–4. GNNs [16] sized to fit one A100 80 GB; desc. = per-neuron descriptor.

One rollout links walking, pausing and turning



Rollout 0–768 ms; click to play at 10× slow motion. Activity panel: [1] Fig. 5.

Latent responses differ across neuron classes

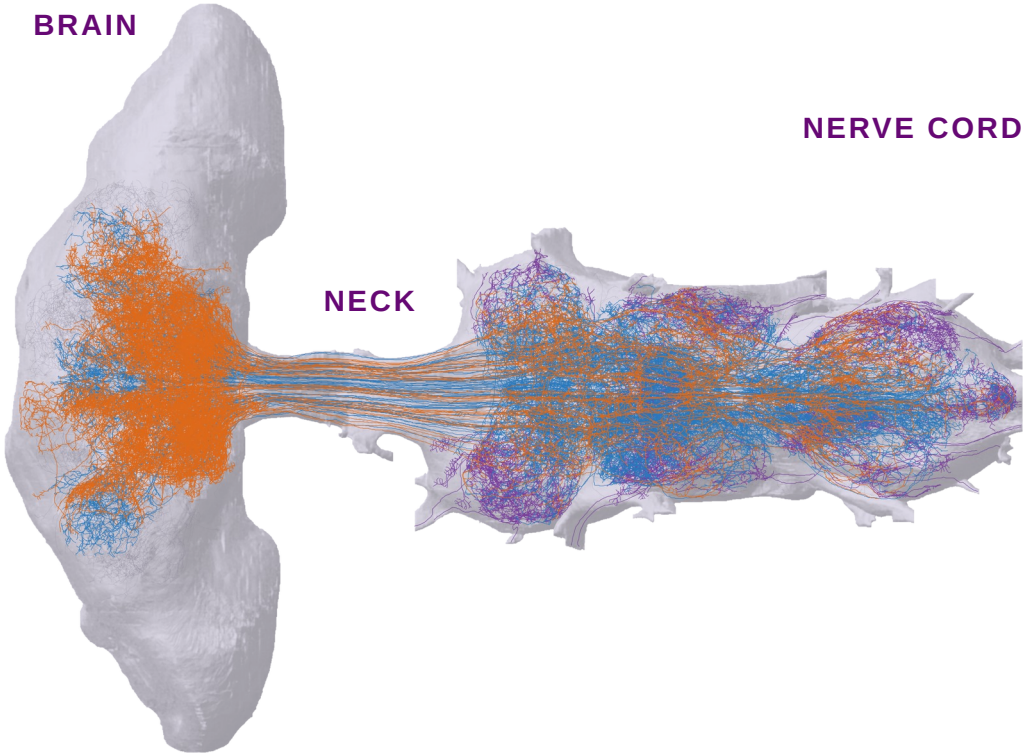


Qualitative latent structure; physiological correspondence remains untested

[1] Fig. 5, Appendix F; random-graph control: Fig. 8. Cf. [17, 18].

OUTLOOK

Next: from brain to brain + nerve cord



166,700

neurons: brain + nerve cord [8]

1,314 down · 1,846 up the neck

WHAT IT ADDS TO FLYGM

- Real motor circuits below the brain
- Body feedback via real pathways
- Updated synapses and cell types [19, 20]

Next tests: longer tasks, recovery, navigation

Outlook, not a result. Rendered from FlyEM male CNS v1.0: shells + ~280 sampled neurons [8].

TAKEAWAYS

A useful prior for whole-body control

1

Wiring as architecture

The fly connectome becomes a whole-body controller.

2

Real wiring matters

Learns faster and steers better than rewired, random and MLP controls.

3

Structured dynamics

Neuron classes show distinct activity patterns (qualitative).

Not an upload: a testable, connectome-structured controller.

Open: biological fidelity · learning without demonstrations · other bodies

[1]. All comparisons use the shared IL + PPO pipeline; the representation analysis is qualitative.

Thank you

Questions are welcome.

SEE THE DEMOS

lnsgroup.cc/research/FlyGM

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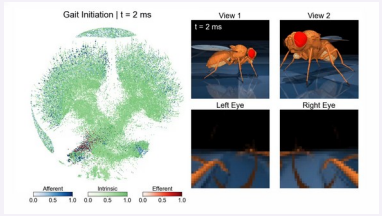
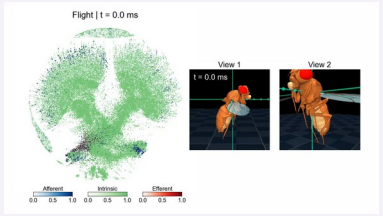
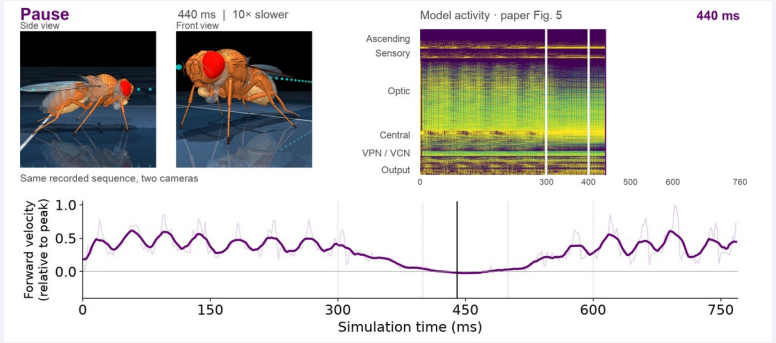
FROM THE DEMO VIDEOS



Walking



Flight



Walk → pause → turn, with model activity

BACKUP

References and resources

DEMO PAGE	PAPER	CONNECTOME DATA	BODY SIMULATOR
lnsgroup.cc/research/FlyGM	arXiv:2602.17997	codex.flywire.ai · male-cns.janelia.org	github.com/TuragaLab/flybody

[1] Z. Jin, Y. Zhu, C. Zhang, Y. Sui. Whole-brain connectomic graph model enables whole-body locomotion control in fruit fly. AIAS+ 2026; arXiv:2602.17997.

[2] S. Dorkenwald et al. Neuronal wiring diagram of an adult brain. Nature 634, 124–138 (2024).

[3] P. Schlegel et al. Whole-brain annotation and multi-connectome cell typing of Drosophila. Nature 634, 139–152 (2024).

[4] P. K. Shiu et al. A Drosophila computational brain model reveals sensorimotor processing. Nature 634, 210–219 (2024).

[5] J. K. Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature 634, 1132–1140 (2024).

[6] S. Wang-Chen et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult Drosophila. Nat. Methods 21, 2353–2362 (2024).

[7] R. Vaxenburg et al. Whole-body physics simulation of fruit fly locomotion. Nature 643, 1312–1320 (2025).

[8] S. Berg et al. Sexual dimorphism in the complete connectome of the Drosophila male central nervous system. Cell 189, 5504–5526 (2026).

[9] Eon Systems. “We’ve uploaded a fruit fly”; “How the Eon team produced a virtual embodied fly”. eon.systems/updates (Mar 2026).

BACKUP

References (continued)

[10] A. Zeleznikow-Johnston. No, we haven’t uploaded a fly yet. LessWrong (Mar 2026).

[11] P. Mineault. Deconstructing viral fly sims. NeuroAI, neuroai.science (2026).

[12] D. A. Pospisil et al. The fly connectome reveals a path to the effectome. Nature 634, 201–209 (2024).

[13] N. Eckstein et al. Neurotransmitter classification from electron microscopy images at synaptic sites in Drosophila melanogaster. Cell 187, 2574–2594 (2024).

[14] E. Todorov, T. Erez, Y. Tassa. MuJoCo: a physics engine for model-based control. IROS, 5026–5033 (2012).

[15] J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347 (2017).

[16] GNN baselines: GCN, Kipf & Welling, ICLR (2017); EdgeConv, Wang et al., ACM TOG 38 (2019); GraphSAGE, Hamilton et al., NeurIPS (2017); GAT, Veličković et al., ICLR (2018); PNA, Corso et al., NeurIPS (2020).

[17] B. E. Brezovec et al. Mapping the neural dynamics of locomotion across the Drosophila brain. Curr. Biol. 34, 710–726 (2024).

[18] E. S. Schaffer et al. The spatial and temporal structure of neural activity across the fly brain. Nat. Commun. 14, 5572 (2023).

[19] S.-C. Yu et al. New synapse detection in the whole-brain connectome of Drosophila. bioRxiv (2025).

[20] D. Deutsch et al. Sexually-dimorphic neurons in the Drosophila whole-brain connectome. Research Square (2025).